

ORACLE

Cloud Operations for Exadata Database Service on Exascale Infrastructure

Database Backup & Recovery

Xxx | xxx

Office of CTO | EMEA Technology Engineering
November 6th, 2025

Agenda

- Introduction
- Backup & Recovery options
 - Oracle Managed Backups
 - Autonomous Recovery Service
 - User Configured Backups
 - Backups using RMAN

Introduction

Introduction



This Workshop provides information about Exadata Database Service on Exascale Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VC, OH, DBs, DG, scaling and patching

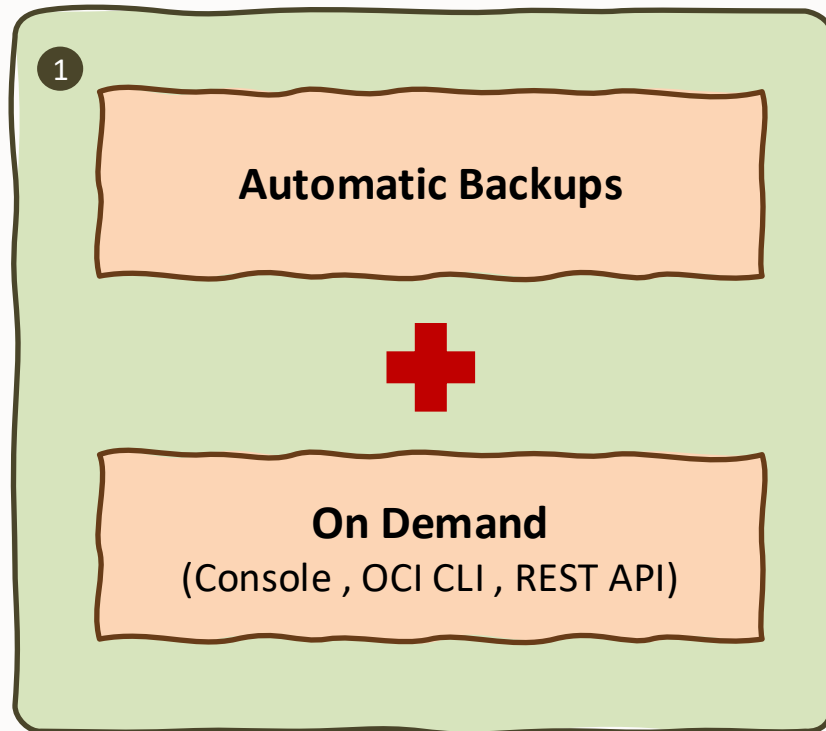
The code in this Workshop is just sample code and can be incomplete.

To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

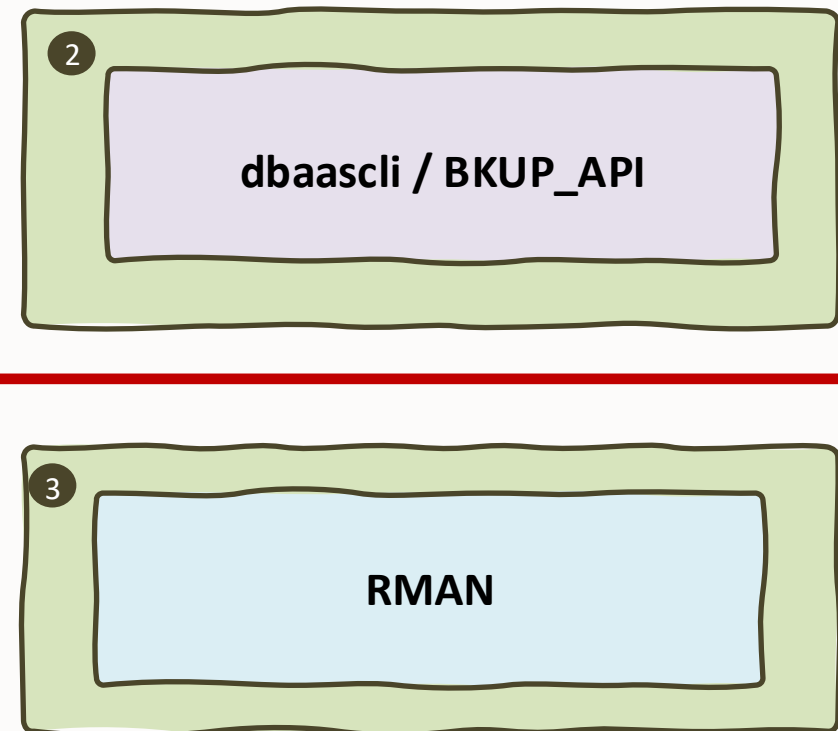
Backup & Recovery options

Options

Oracle Managed Backups

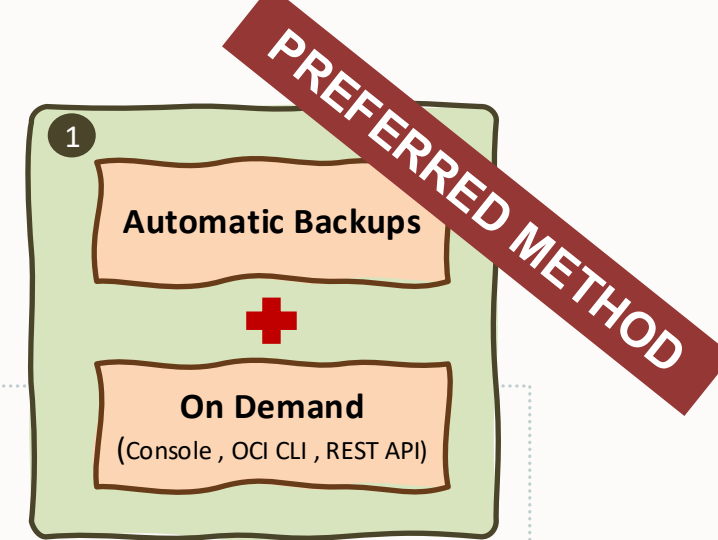


User Configured Backups



Oracle Managed Backups

Automatic Exadata Database Backups



- Managed by Oracle Cloud Infrastructure
- One time configuration at database creation, or later, using the Console, OCICLI or the API.
- Incremental backup strategy.
- Use the Console, OCICLI and APIs to list the managed backups.
- Backup deletion on database termination
- All backups are encrypted with the same master key used for Transparent Data Encryption (TDE) wallet encryption.
- A common administrative user **C##DBLCMUSER** with SYSBACKUP role at the CDB\$ROOT level is created.
 - Credentials for this user are randomly generated and securely managed by cloud automation.
 - If the user is not found or is LOCKED and EXPIRED, then cloud automation will recreate or unlock this user during the backup or recovery operation.

Exceptions

- The **dbaascli database backup** and **dbaascli database recover** commands can be used for certain operations.
- Customers are allowed to query RMAN views or issue RMAN restore and recovery commands, for example, table, datafile, or tablespace recovery commands.
- **Do not use RMAN** configuration to change any of the pre-tuned cloud RMAN settings.

Automatic Exadata Database Backups

Autonomous Recovery Service

- Fully managed service
- Incremental forever strategy
- Based on the on-premises Oracle's ZDLRA technology
 - Protect changes in real time
 - Validate backups without production database overhead
 - Fast, predictable recovery to any point in time
- Autonomous Recovery Service protection policy
 - Bronze :14 days
 - Silver: 35 days - Default
 - Gold: 65 days
 - Platinum: 95 days
 - Custom
- Requirements
 - Create IAM policies
 - Configure network resources
 - Register a Recovery Service subnet
 - Create protection policies
- Backup deleted on database termination according the deletion option selected.

Oracle Object Storage

- A secure, scalable, on-demand storage solution
- L0 (once/week) + daily L1
- The retention periods (in days) are 7, 15, **30(Def)**, 45, 60.
- Automatic backups are deleted upon database termination.
 - Standalone backups will remain after database termination.
 - Must be manually deleted.
 - Standalone backups can be used to create a new database.
- Requirements
 - An existing Object Storage bucket
 - Access to the OCI Object Storage
 - Recommended: Service Gateway
 - An auth token
 - User with tenancy-level access to Object Storage

- ✓ **Default** backup window of **00:00 to 06:00**
- ✓ **Optional 2-hour** daily backup window.
 - 12 scheduling windows to choose.
 - Starting on an even-numbered hour.
- ✓ The automatic backup process starts at **any time within** the assigned window.
- ✓ Backups jobs **do not** necessarily complete within the scheduling window.

Default Backup Channel Allocation

- These are the default settings when using "Oracle Managed Backup" or "User Configured Backup".
- The number of channels is determined:
 - At the time the backup or restore command is executed.
 - By the core count of the node.
- Restore operations are prioritized.
- The cluster-wide total channel count is the per node value multiplied by the number of nodes.
- The automation uses the SCAN to distribute RMAN channels across all nodes in the cluster.

Cores Per Node	Backup Channels/Node	Restore Channels/Node
Cores <=12	2	4
Cores >12 and Cores <=24	4	8
Cores >24	8	16

✓ A static per node **bkup_channels_node** value can be set by using the **DBAASCLI getConfig/configure** to generate a **bckup cfg** file.

- Valid values are 1 - 32
- **Total = bkup_channels_node * #nodes**
- **Limit** of 255 channels.



Long-Term Retention Backup with Recovery Service

Long-term retention backup (LTR) allows you to store full backups for periods up to ten years for compliance, regulatory, or other business needs with complete LTR lifecycle management and immutability.

- Retention period must be in Days (90 - 3,650) or years (1 - 10) from when the backup was created.
- Does not require creating a new full production backup.
- Retention period can be changed within the retention period.
- You can restore an LTR backup to create a new database within the retention period.
- Upon terminating a database, the LTR backups will be deleted:
 - In 72 hours
 - On policy (Recommended)

Other considerations

- Will exist independently of any automatic backups and be automatically deleted after the specified retention period ends.
- On Data Guard configuration, LTR will be created only for the database where it is requested.
- The database must be in the AVAILABLE state.
- Supported for databases with file-based TDE or KMS-based keystores. Encryption keys will be maintained for the entire retention period.
- Can be canceled while it is in the 'creating' state.
- Can be deleted at any time after it is created.
- In-place restore is not supported for LTR. During restore:
 - If the backup is of a supported DBHome major version, it will be restored to the latest RU of that version.
 - If the backup is of an unsupported DBHome major version, it will be restored to a supported major version, database must be upgraded.

Manage Automatic Backups using the Console

Enable backups in an existing database

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Container databases → Select the Container database → More actions → **Configure automatic backups**

Check backups

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Container databases → Select the Container database → **Backups**

Create an on-demand backup

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Container databases → Select the Container database → Backups → **Create backup**

Create a database from backup

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Select VM Cluster →
Container databases → Select the Container database → Backups → Select Actions (3 dots) on the backup → **Create database**

Delete an on-demand backup

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Select VM Cluster →
Container databases → Select the Container database → Backups → Select Actions (3 dots) on the backup → **Delete**

Manage Automatic Backups using APIs

Activate backups during database creation

```
POST /20160918/databases
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "dbHomeId" : "ocidl.dbhome.oc1.phx.<unique_ID>",
  "source": "NONE",
  "database" : {
    "adminPassword" : "<password>",
    "dbName" : "myTestDb",
    "dbBackupConfig" : {
      "recoveryWindowInDays" : 30,
      "autoBackupEnabled" : true
    }
  },
  "kmsKeyId" : "ocidl.key.oc1..<unique_ID>",
  "kmsKeyVersionId" : "ocidl.keyversion.oc1..<unique_ID>"
}
```

Lists backups

```
GET /20160918/databases/backups?databaseId=<databaseId>&limit=10
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Get backup information

```
GET /20160918/backups/<backupId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Update backup parameters

```
PUT /latest/databases/<databaseId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "dbBackupConfig" :
  {
    "recoveryWindowInDays" : 30,
    "autoBackupEnabled" : true
  }
}
```

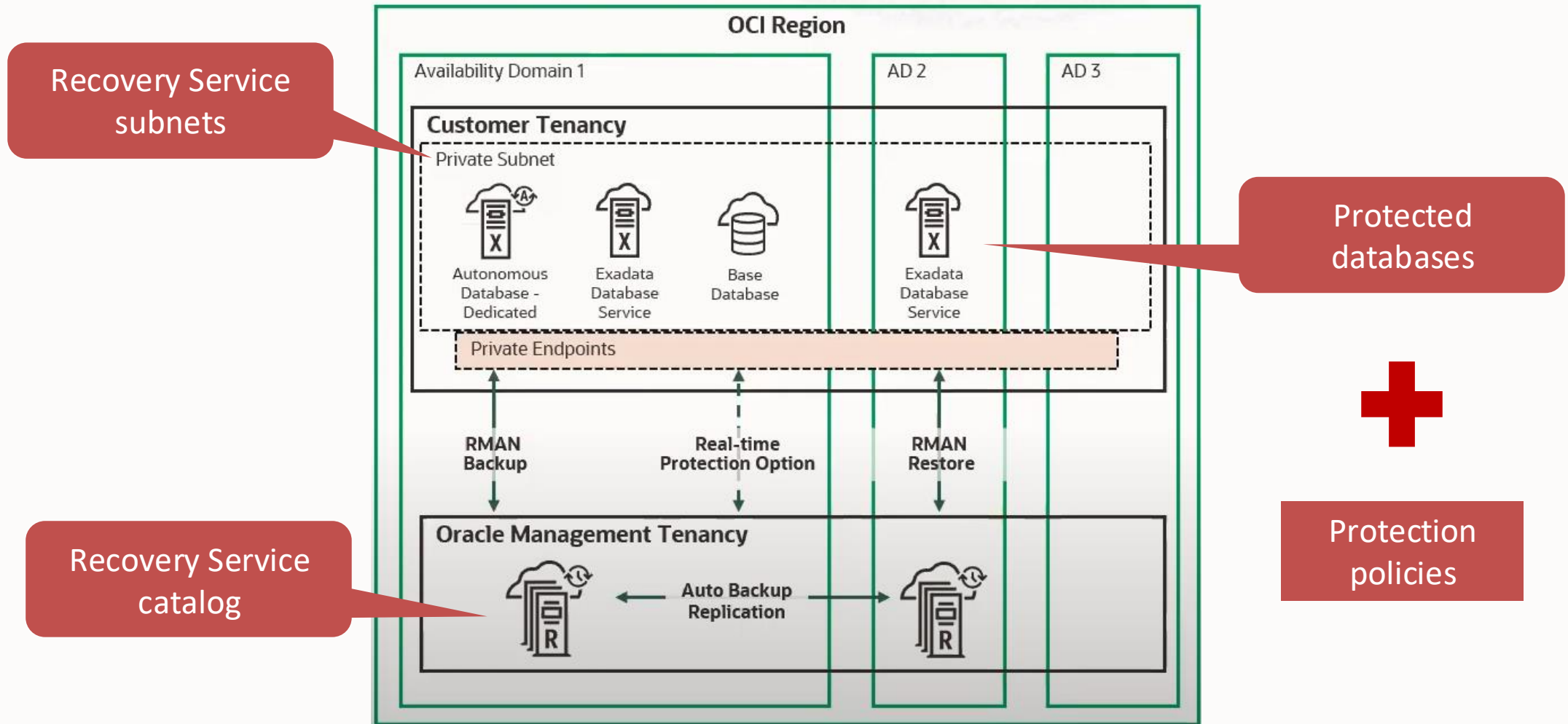
Delete a full backup

```
DELETE /20160918/backups/<backupId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Autonomous Recovery Service

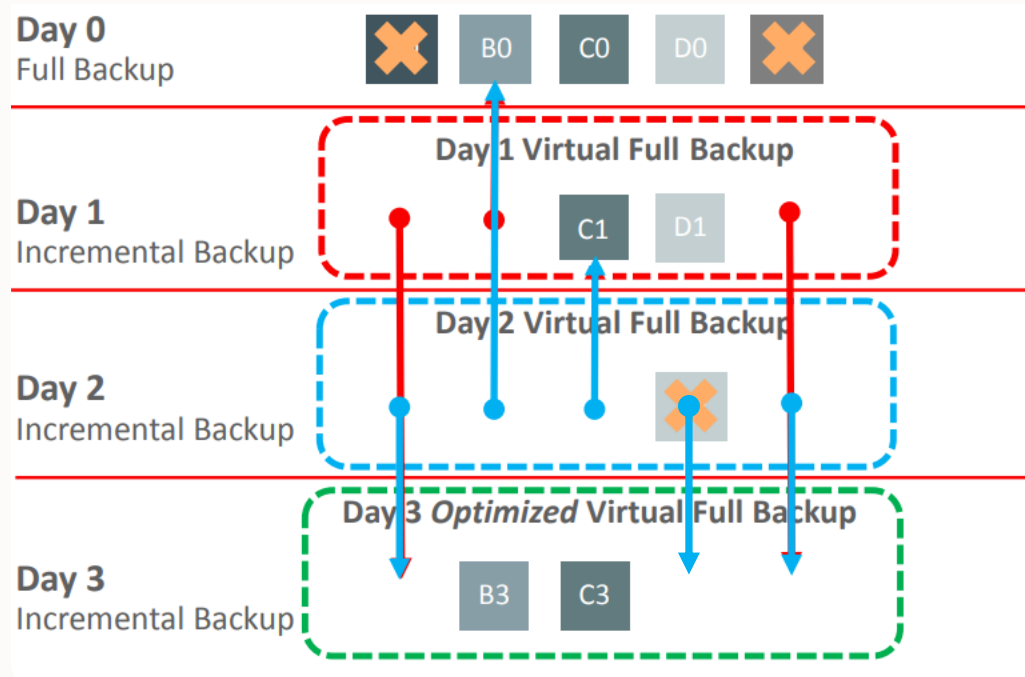
Autonomous Recovery Service

Components



Backup strategy

Delta Store



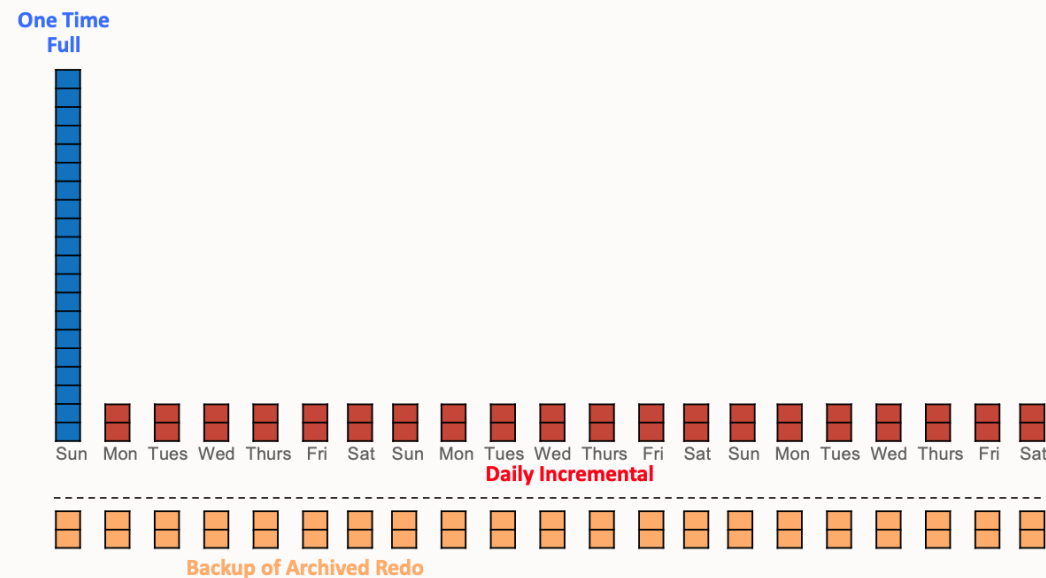
- The backup strategy starts with an initial full (INCREMENTAL LEVEL 0) backup.
- After the LEVEL 0 backup, only incremental backups are needed.
- A virtual full backup is created in the Recovery Appliance catalog for each received RMAN incremental backup.
- It operates at a data file level and appears as a normal LEVEL 0 backup in RMAN LIST/REPORT & catalog queries.
- OPTIMIZE task is a background process that runs on the most recent virtual full backup to 'optimize' its read I/O access for future restore operations.
- Virtual full blocks are reordered into contiguous sets for fewer & larger read I/O requests, which is more efficient than many smaller read I/O requests.

Backup strategy

RCV vs Object Storage

Autonomous Recovery Service

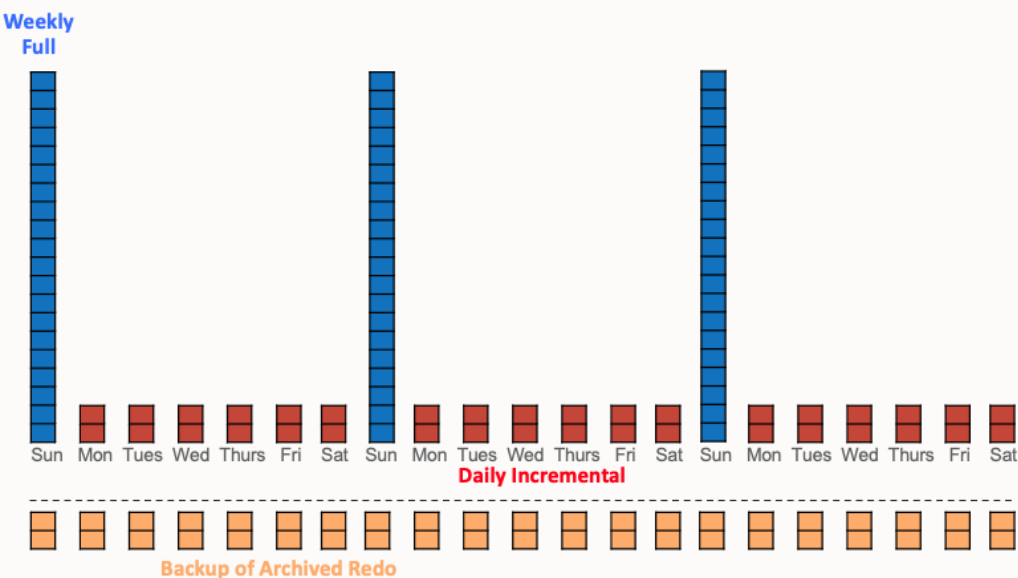
Incremental forever with full backup virtualization



Benefit for operations & cost:
Eliminating the repeated full reduces backup time & lowers resources, which saves time & money

Object Storage

Weekly full with daily incremental



Weekly full backups move a lot of repeat data & require more time & resources

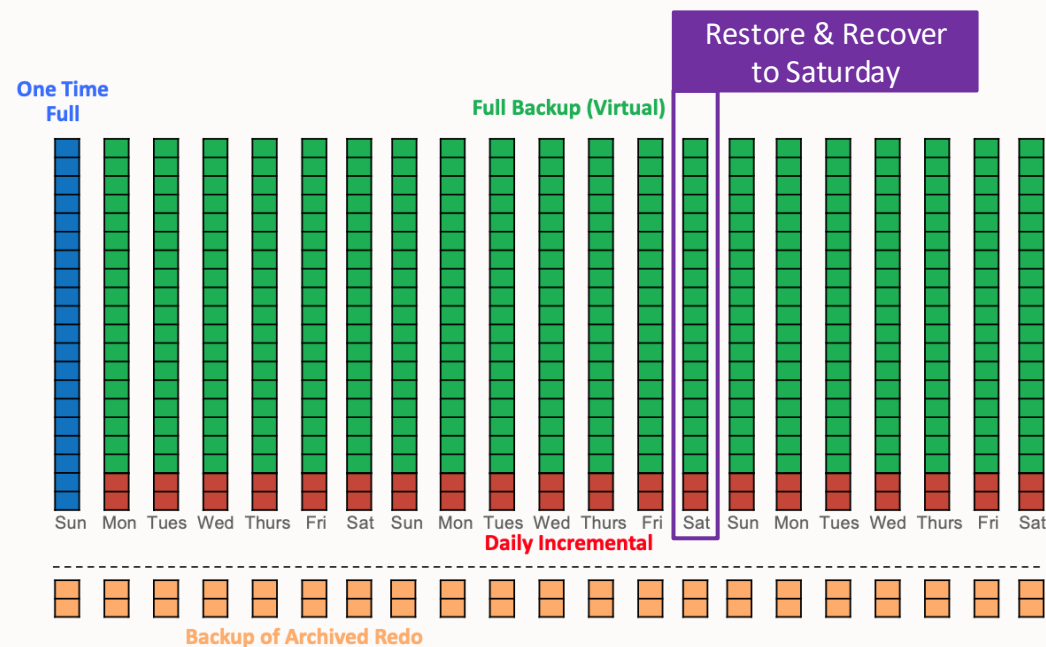


Backup strategy

RCV vs Object Storage

Autonomous Recovery Service

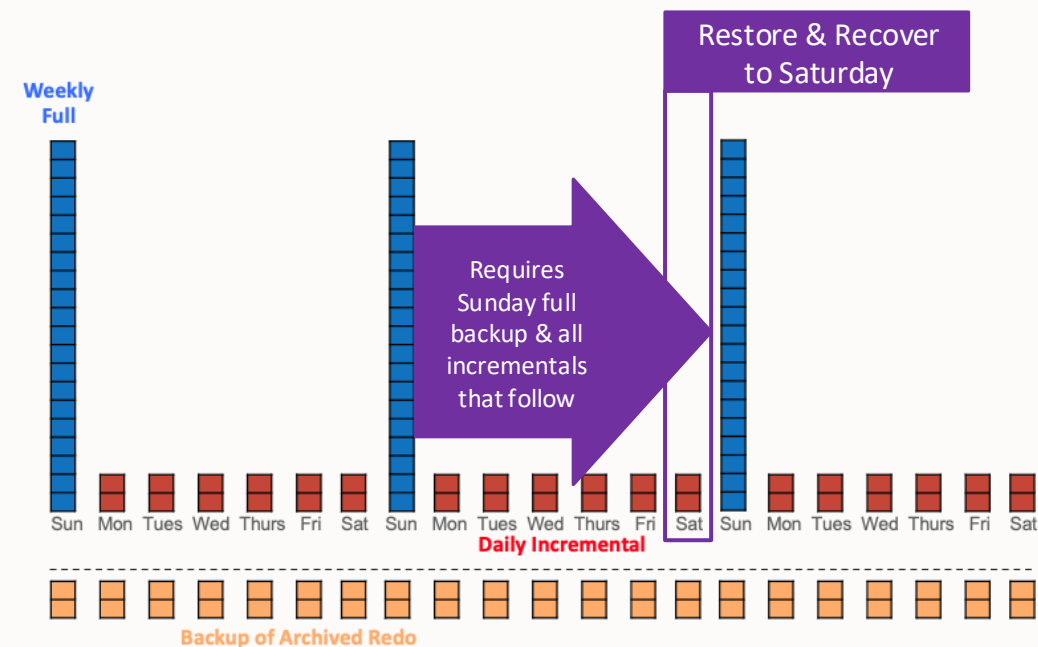
Every incremental backup becomes a virtual full backup



Benefit for operations:
Predictably fast restore with no extra incremental backups to process.

Object Storage

Every incremental relies on the previous full & incremental



Restore time is less predictable since each incremental backup after the full must be processed. Sun, Mon, Tues, Wed, Thurs & Fri must be processed.

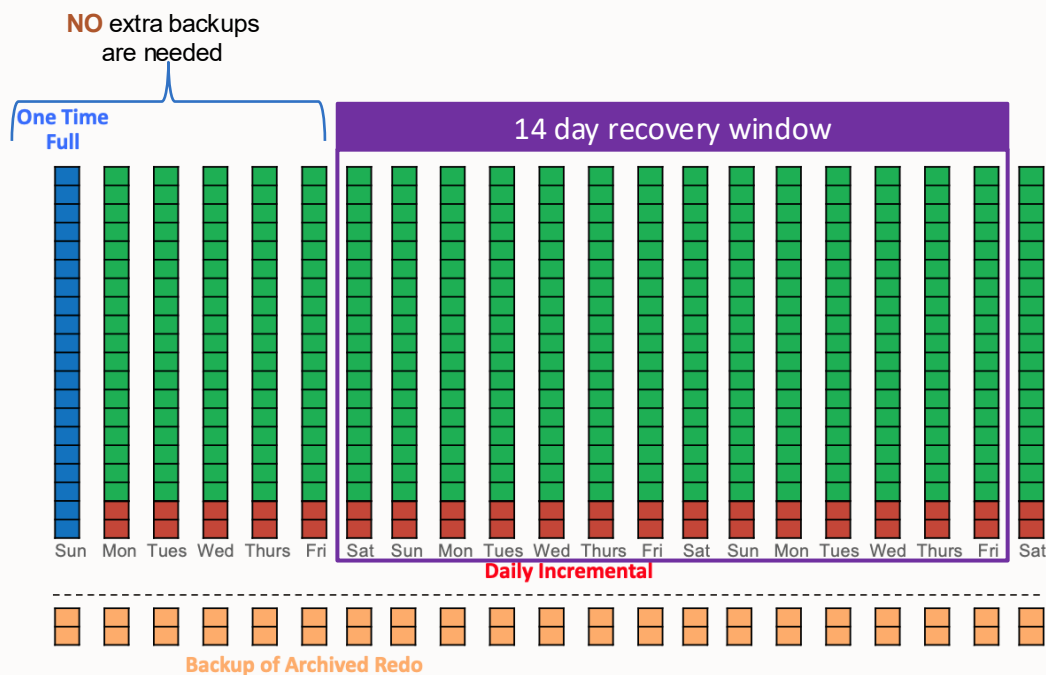


Backup strategy

14 days recovery window

Autonomous Recovery Service

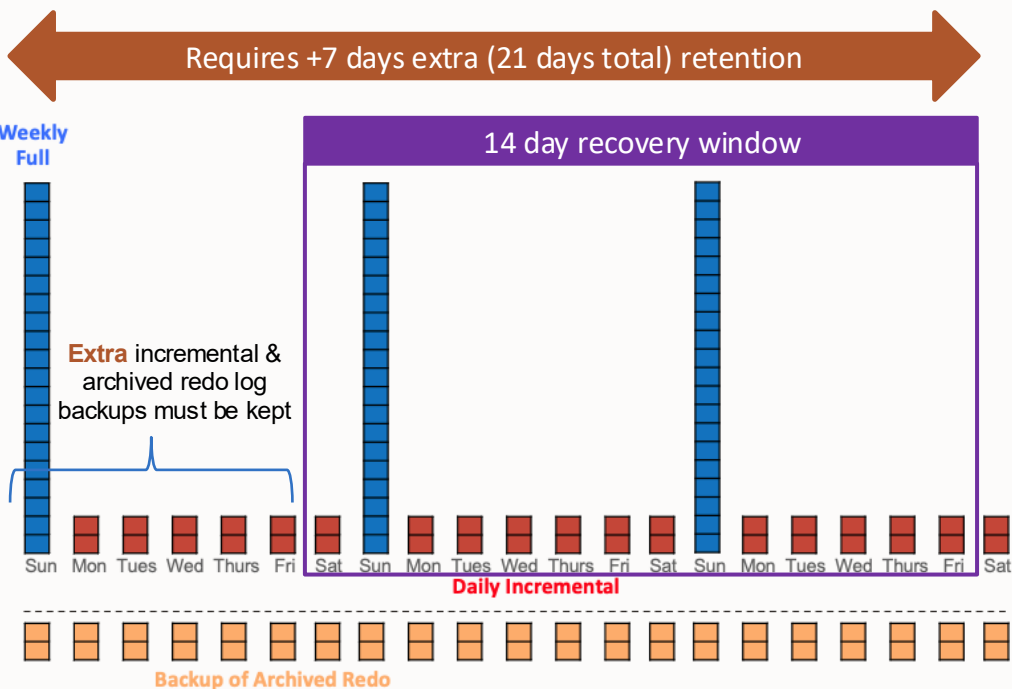
Each daily backup can be purged independently



Benefit for operations & cost:
Only keep the backups needed, since there is no interdependency between database backups.

Object Storage

Extra backups must be retained for recovery



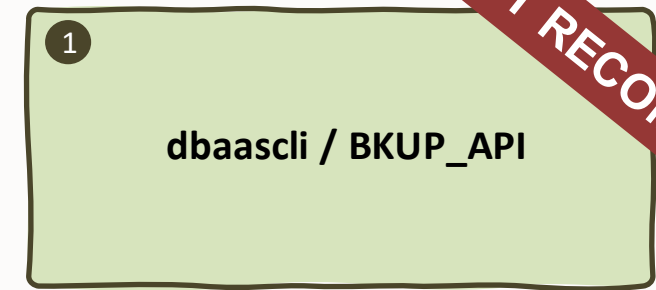
More data has to be retained since backups are interdependent. Saturday can't be restored without the full backup from the previous Sunday and the Mon – Fri incremental and archived redo log backups.



Customized Backups

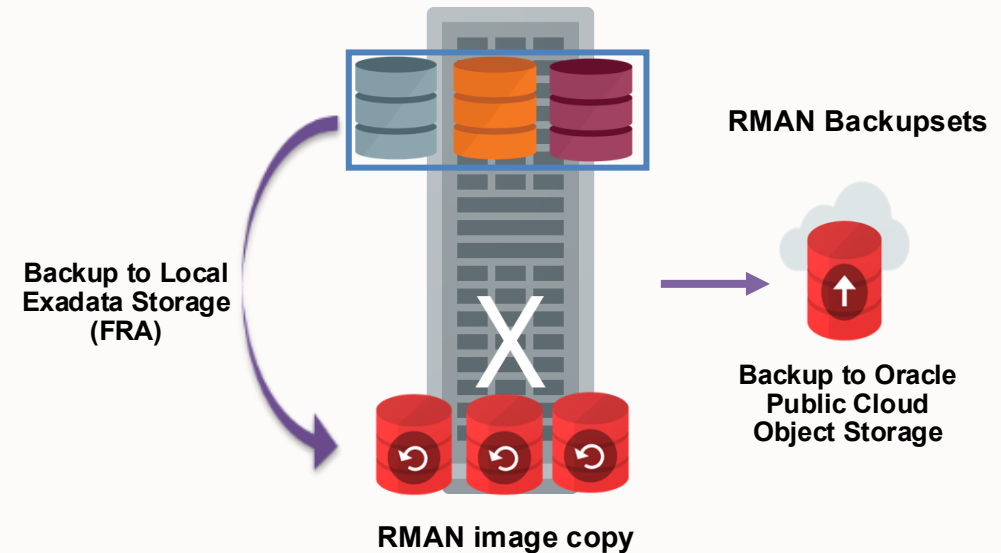
Customized Backups using dbaascli

- Customer managed.
- Configured using Local VM Command Line dbaascli.
- Not synchronized with the Control Plane nor integrated with the OCI APIs.
- Neither management nor lifecycle operations on these backups are supported from the service Control Plane console.
- Use cases:
 - Direct access to Backup destinations is required.
- Customer must take care of:
 - TDE wallet backups
 - Archive logs cleanup jobs
- **Do not use RMAN** backups in conjunction User Configured Backups.
- All backups are encrypted.



NOT RECOMMENDED

Destinations



Backup Setup



1. Connect to one VM cluster node and become root

```
> ssh -i <private_key_path> opc@<node_1_ip_address>
$ sudo su -
```

2. Generate a file containing the current backup settings for the database deployment

```
# dbaascli database backup --getConfig [--configFile <file_name>] --dbname <database_name>
```

3. Modify the parameters in the file to meet your requirements

4. Associate this backup configuration with a database name

```
# dbaascli database backup --configure --configFile <file_name> --dbname <database_name>
```

5. Check the status of UUID generated for this command

```
# dbaascli database backup --status --uuid <uuid> --dbname <database_name>
```

To remove the configuration:

Repeat the procedure using a temp file with the following parameters:

- bkup_oss=no
- bkup_cron_entry=no
- bkup_archlog_cron_entry=no

Backup Configuration Parameters

PARAMETER	OLD NAME	DESCRIPTION
<code>backupDestination=oss</code>	<code>bkup_type</code>	Enables backups to cloud storage.
<code>bkup_oss=[yes no]</code>		Whether to back up to Object Storage.
<code>ossURL¹=<swift_url></code>	<code>bkup_oss_url</code>	The Object Storage URL including the tenant and bucket you want to use.
<code>ossUserName¹</code>	<code>bkup_oss_user</code>	The user name for the Oracle Cloud Infrastructure user account. Must be a member of the Administrators group.
<code>ossAuthToken¹</code>	<code>bkup_oss_passwd</code>	The auth token generated by using the Console or IAM API, as described in Prerequisites.
<code>bkup_oss_recovery_window¹=[7-90]</code>		The number of days for which backups and archived redo logs are maintained in the Object Storage bucket.
<code>ossRecoveryWindow²=[1-90] (Def:30)</code>	<code>bkup_oss_recovery_window</code>	Retention period (days) for backups to cloud storage.
<code>bkup_archlog_cron_entry=[yes no]</code>		When no backups are configured using dbaastools, setting <code>bkup_archlog_cron_entry=no</code> will remove the archive log clean up job from crontab. The default value is “yes”.
<code>bkup_daily_time=hh:mm</code>		The time at which the daily backup is scheduled, specified in hours and minutes (hh:mm), in 24-hour format.



Backup Configuration Parameters

PARAMETER	OLD NAME	DESCRIPTION
<code>compressionLevel = [NONE basic low medium high]</code>	<code>bkup_rman_compression</code>	Level of compression applied to automatic backups.
<code>sectionSize (Def: 64G)</code>	<code>bkup_section_size</code>	RMAN section size that is used for automatic backups.
<code>channelsPerNode = [1-32] (Def:2)</code>	<code>bkup_channels_node</code>	Number of RMAN channels per node used for automatic backups.
<code>autoBackupTime</code>	<code>bkup_daily_time</code>	Start time of the automatic daily backup expressed in 24-hour time as hh:mm.
<code>backupFrequencyAL=[15, 20, 30, 60, 120 through 1440 in one-hour intervals expressed in minutes]</code>	<code>bkup_archlog_frequency</code>	Interval in minutes between automatic backups of archived database log files.
<code>filesPerSet</code>	<code>bkup_filesperset_regular</code>	Specifies the maximum number of data files that can be included in a backup set for Regular/Archival backups.
<code>filesPerSetAL</code>	<code>bkup_filesperset_al</code>	Specifies the maximum number of archive log files that can be included in a backup set for Archivelog Backups.
<code>Encryption</code>	<code>bkup_encryption</code>	Encryption specifies whether backups should be encrypted or not.
<code>optimization</code>	<code>rmanBackupOptimization</code>	Optimization is a feature in that reduces the amount of data that needs to be backed up, transferred, and restored. Recommended value is ON.
<code>compressionAL</code>	<code>Compress_Archive_Logs</code>	Specifies whether to compress the archive log backups are not. Not applicable to Recovery Service.



Backup Configuration Parameters

PARAMETER	OLD NAME	DESCRIPTION
<code>bkup_disk=[yes no]</code>		Whether to back up locally to disk (Fast Recovery Area).
<code>numberOfChannelsForFraCleanup</code>	<code>rmanFraCleanupChannels</code>	Specifies the number of channels used for FRA Cleanup job.
<code>archivelogRetentionDays</code>	<code>bkup_archlog_fra_retention</code>	Specifies the number of days archive log to be retained in FRA.
<code>scheduleBackups = [yes no]</code>	<code>bkup_cron_entry</code>	Enables the automatic backup configuration.
<code>manageArchivelogs = [yes no]</code>	<code>bkup_archlog_cron_entry</code>	Enables automatic backups of archived database log files.
<code>L0BackupDay = [mon tue wed thu fri sat sun]</code>	<code>bkup_l0_day</code>	Day of the week when a level 0 backup is taken.
<code>useCatalog¹ = [yes no]</code>	<code>bkup_use_rcat</code>	Enables the use of an existing RMAN recovery catalog.
<code>catalogUserName¹</code>	<code>bkup_rcat_user</code>	Recovery catalog user name.1
<code>catalogPassword¹</code>	<code>bkup_rcat_passwd</code>	Password for recovery catalog user specified in <code>bkup_rcat_user1</code>
<code>catalogConnectionString¹</code>	<code>bkup_rcat_conn</code>	Connection string for the RMAN recovery catalog.1

¹ Applicable only when `bkup_use_rcat` is set to yes.



Local VM Command-Line Interfaces - dbaascli



Backup CDB related operations

```
$ dbaascli database backup --dbname <value>1
```

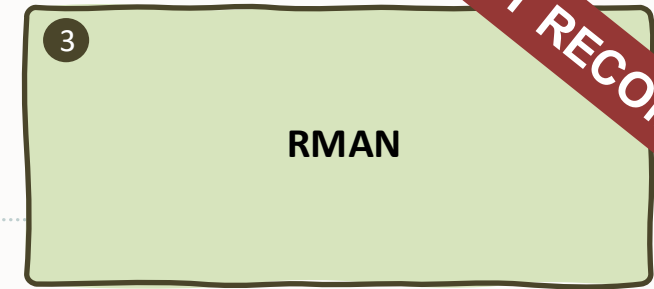
Backup PDB

```
$ dbaascli pdb backup --dbname <value> --pdbName <value>1
```

¹... Not all parameters listed. Please check the command definition

Oracle Managed Backups

Customized Backups using RMAN



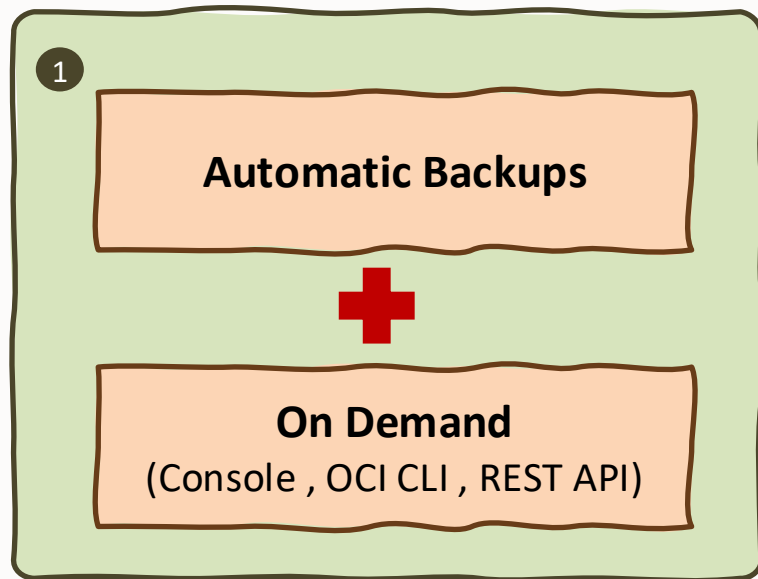
- Backups can be directly taken using RMAN with customer-owned customized scripts.
- If you plan to backup using RMAN, then you must unregister the database from backup automation.
- Incompatible with the other 2 backups methods.
- Managing backup and recovery, using RMAN, on Exadata Cloud Infrastructure requires taking full ownership of both database and archive log backups, and the cloud-managed backups should no longer be used. Before manual backups are started, the cloud-managed backup functionality should be disabled.
- Replicas:
 - RMAN catalog possible.
 - Set up backup replication.
 - Not compatible with the two previous options, so no cloud automation. Mix options could result in conflicting conditions or over-writes of settings, and backups may not execute successfully.
- Only leverage if long term (archival) backup retention, backup to remote region or offload backup to standby use cases required.

Who can use this option?

- Customers who want to maintain their existing RMAN backup/restore scripts.
- Customers who want to configure backups from Standby database in Data Guard environments to offload the backup workload to Standby.

Restore Options

Restoring Managed Backups



- Restore options: Latest, timestamp or scn.
- In-place CDB restore.

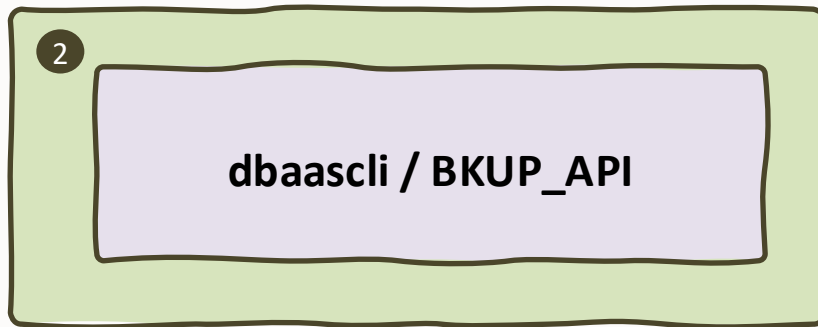
Restore a database from backup

Menu → Oracle AI Database →
Oracle Exadata Database Service on Exascale Infrastructure →
VM Clusters → Select VM Cluster →
Container databases → Select the Container database →
More actions → **Restore**

Restore a backup using Rest APIs

```
POST /20160918/databases/<databaseId>/actions/restore
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "restoreDetails" :
  {
    "databaseSCN" : null,
    "timestamp" : null,
    "latest" : true
  }
}
```

Restoring User Configured and RMAN Backups



- Restore CDB and PDB capabilities.
- No duplicate on the same host via bkup_api

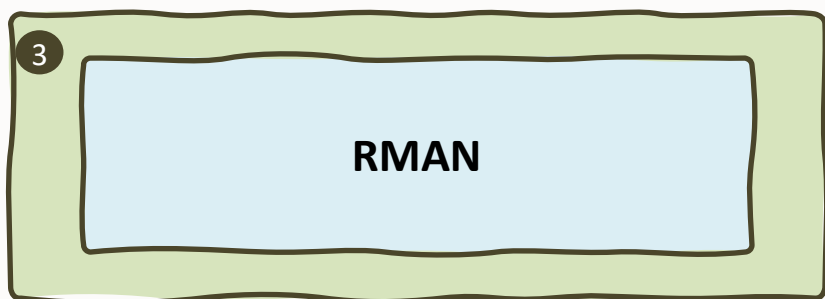
Restore CDB

```
$ dbaascli database recover --dbname <value>  
  --start [--untilTime <value> [--nonUTC] | --untilSCN <value> | --latest | --tag <value>]1
```

Restore PDB

```
$ dbaascli pdb recover --dbname <value> --pdbName <value>  
  --start [--untilTime <value> [--nonUTC] | --untilSCN <value> | --latest | --tag <value>]1
```

Restoring RMAN Backups

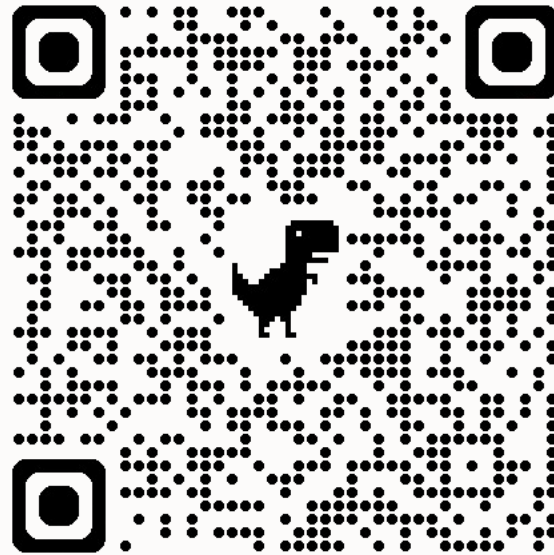


- Anywhere the backups reside (local OSS bucket, remote bucket across AD, remote bucket across region)

RMAN scripts containing

```
$ restore database;  
$ recover database;
```


Stay tuned



Operations Advisory
GitHub Repository



Thank you

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